Project 4: Cryptography MATH 4175
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Part 1: Difference Distribution Table (ND(a',b'))
Rows: Input Difference, Columns: Output Difference

Input Diff/ Output Diff	0	1	2	3	4	5	6	7
0	8	0	0	0	0	0	0	0
1	0	4	0	4	0	0	0	0
2	0	0	0	0	2	2	2	2
3	0	0	0	0	2	2	2	2
4	0	0	0	0	2	2	2	2
5	0	0	0	0	2	2	2	2
6	0	4	4	0	0	0	0	0
7	0	0	4	4	0	0	0	0

Parts 2 & 3: Propagation Ratios

Trail	Ratio (fraction)	Ratio (decimal)
Tr1 (P6 -> H6)	1/4	0.25
Tr2 (P6 -> H5,H6)	1/8	0.125
Tr3 (P6 -> H4,H5,H6)	1/8	0.125

Part 4: Filtering for Right 4-Tuples

All 4-tuples with y^y* values:

x	x*	y	y*	y^y*
100111	100110	100100	111110	011010
000111	000110	110010	110110	000100
001100	001101	111001	100000	011001
011000	011001	011101	011111	000010
001000	001001	001101	000011	001110
011010	011011	101001	101000	000001

Right 4-tuples:

x	x*	y	y*
000111	000110	110010	110110

011000	011001	011101	011111
011010	011011	101001	101000

Part 5: Weighted Cryptanalysis

(a) Counter ca for Trail Tr1:

Key	Binary	ca
0	000	1
1	001	1
2	010	1
3	011	1
4	100	0
5	101	0
6	110	0
7	111	0

(b) Counter cb for Trail Tr2:

Key	Binary	cb
0	000	0
1	001	0
2	010	0
3	011	1
4	100	0
5	101	0
6	110	0
7	111	1

(c) Counter cc for Trail Tr3:

Key	Binary	cc
0	000	1
1	001	1
2	010	1
3	011	1
4	100	0
5	101	0

6	110	0
7	111	0

(d) Weighted Average Counts C:

Key	Binary	c1	c2	c3	Count C
0	000	1	0	1	0.375
1	001	1	0	1	0.375
2	010	1	0	1	0.375
3	011	1	1	1	0.5
4	100	0	0	0	0
5	101	0	0	0	0
6	110	0	0	0	0
7	111	0	1	0	0.125

(e) Conclusion:
Last 3 bits of K4: 011